# THE BASE CHANGE WHERE A MAP BECOMES ZERO IS CUT OUT BY A CLOSED SUBSCHEME, OF FINITE PRESENTATION IN THE FINITE TYPE CASE

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: a reduction to affine pieces, a value by value construction of the killing ideal through Lazard's theorem, and a limit argument producing a finitely generated ideal in the finite type case

*Editorial note.* The label lemma-F-zero-somewhat-closed names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Quasi-coherent modules on algebraic spaces, covers by disjoint unions of affine schemes, and the go-up lemma allowing the statement to be checked after a surjective etale base change
- Lazard's theorem that a flat module is a filtered colimit of free modules, and commutation of tensor product with colimits
- The limit criterion that a vanishing statement stable under base change holds already at some stage of a filtered colimit of closed subschemes

*Source reference.* Original theorem 058G: sources/stacks-chow/tags.txt, source lines 6779–6779. Prerequisite pointer invoked at proof line(s) 1604; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 083I: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 1291–1291. Prerequisite pointer invoked at proof line(s) 1582; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 083J: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 1335–1335. Prerequisite pointer invoked at proof line(s) 1622; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0CWA: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 1560–1560. Prerequisite pointer invoked at proof line(s) 1568; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 03FX: sources/stacks-chow/tags.txt, source lines 4508–4508. Prerequisite pointer invoked at proof line(s) 1581; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *In Situation 0CWA. Let $T \to S$ be a quasi-compact morphism of schemes such that the base change $u_T$ is zero. Then exists a closed subscheme $Z \subset S$ such that (a) $T \to S$ factors through $Z$ and (b) the base change $u_Z$ is zero. If $\mathcal{F}$ is a finite type $\mathcal{O}_X$-module and the scheme theoretic support of $\mathcal{F}$ is quasi-compact, then we can take $Z \to S$ of finite presentation.*

*Proof.* Let $U \to X$ be a surjective étale morphism of algebraic spaces where $U = \coprod U_i$ is a disjoint union of affine schemes (see Properties of Spaces, Lemma 03FX). By Lemma 083I we see that we may replace $X$ by $U$. In other words, we may assume that $X = \coprod X_i$ is a disjoint union of affine schemes $X_i$. Suppose that we can prove the lemma for $u_i = u|_{X_i}$. Then we find a closed subscheme $Z_i \subset S$ such that $T \to S$ factors through $Z_i$ and $u_{i,Z_i}$ is zero. If $Z_i = \operatorname{Spec}(R/I_i) \subset \operatorname{Spec}(R) = S$, then taking $Z = \operatorname{Spec}(R/\sum I_i)$ works. Thus we may assume that $X = \operatorname{Spec}(A)$ is affine.

Choose a finite affine open covering $T = T_1 \cup \ldots \cup T_m$. It is clear that we may replace $T$ by $\coprod_{j=1,\ldots,m} T_j$. Hence we may assume $T$ is affine. Say $T = \operatorname{Spec}(R')$. Let $u : M \to N$ be the homomorphisms of $A$-modules corresponding to $u : \mathcal{F} \to \mathcal{G}$. Then $N$ is a flat $R$-module as $\mathcal{G}$ is flat over $S$. The assumption of the lemma means that the composition

$$M \otimes_R R' \to N \otimes_R R'$$

is zero. Let $z \in M$. By Lazard's theorem (Algebra, Theorem 058G) and the fact that $\otimes$ commutes with colimits we can find free $R$-module $F_z$, an element $\tilde{z} \in F_z$, and a map $F_z \to N$ such that $u(z)$ is the image of $\tilde{z}$ and $\tilde{z}$ maps to zero in $F_z \otimes_R R'$. Choose a basis $\{e_{z,\alpha}\}$ of $F_z$ and write $\tilde{z} = \sum f_{z,\alpha} e_{z,\alpha}$ with $f_{z,\alpha} \in R$. Let $I \subset R$ be the ideal generated by the elements $f_{z,\alpha}$ with $z$ ranging over all elements of $M$. By construction $I$ maps to zero in $R'$ and the elements $\tilde{z}$ map to zero in $F_z/IF_z$ whence in $N/IN$. Thus $Z = \operatorname{Spec}(R/I)$ is a solution to the problem in this case.

Assume $\mathcal{F}$ is of finite type with quasi-compact scheme theoretic support. Write $Z = \operatorname{Spec}(R/I)$. Write $I = \bigcup I_\lambda$ as a filtered union of finitely generated ideals. Set $Z_\lambda = \operatorname{Spec}(R/I_\lambda)$, so $Z = \operatorname{colim} Z_\lambda$. Since $u_Z$ is zero, we see that $u_{Z_\lambda}$ is zero for some $\lambda$ by Lemma 083J. This finishes the proof of the lemma. □